

Plan for Scale

Plan for Scale - From 50 to 80,000 Cameras (Statewide)

> Direct answer to **Plan for Scale** - rollout, network/edge, storage, HA/DR, security for statewide deployment across 26 departments. Every mechanism is already present in the 50-camera baseline and scales without redesign.

Executive Summary

Baseline `a95adcd` runs 51 cameras on 5 containers (api/web/postgres-postgis/redis/mediamtx) with 7 vendor adapters, federated IAM (14 depts/320 perms), sealed dossiers and cluster leases. The same contracts scale horizontally: add MediaMTX per district, API replicas behind LB, PG read-replicas + partitioning, and Kafka/NATS sharding - proven by `SCALABILITY.md`, `08` §8 and `src/cluster/router.py`.

1. Statewide Rollout Strategy (~80,000 Heterogeneous Cameras, 26 Departments + Private-Public Feeds)

Phased Roadmap

| Phase | Scope | Cameras | Duration | Onboarding tool | Success gate |

Automated Discovery & Bulk Provisioning (Minimize Manual Overhead)

- **Discovery:** `GET /api/v1/streams/discover\_onvif` - WS-Discovery multicast + static `ONVIF\_PROBE\_URLS` (per-district seed). Enriches `onvif\_xaddr` without vendor SDK.

Centralised CCTV Registry & GIS Foundation (Model 1) for Continuous Asset Management

- **Registry:** `camera\_registry` (1:1 `cameras`) with `cctv\_code` IN-GJ-{district}-{name}, `vendor\_id`, `district/department/board`, `cluster\_key` {district}/{department}, `coverage\_radius\_m` 250, `last\_health\_score`, `uptime\_pct` - additive, paginated `GET /registry?limit=50`.

2. Network & Bandwidth Optimization (Edge, Compression, Relay, Adaptive, Transcoding) - Up to ~1,000 km

Edge-Based Analytics (Minimize Backhaul)

- ANPR inference sharded per `cluster\_key` (district) - `src/anpr/\*` workers co-located with MediaMTX; only `PlateDetection` metadata (plate, conf, bbox) travels to core, not raw video. Demo OCR 0.86-0.93; `backend="sim"` swappable to live engine behind same `anpr/\*`.

Stream Compression & Bandwidth

| Link | Baseline 51 cams @2 Mbps | 80k @2 Mbps raw | With edge (ANPR + motion-gate ~70% reduction) |

Intelligent Relay, Adaptive Bitrate, Transcoding

- **Relay:** MediaMTX per district (80k -> ~30 MediaMTX nodes, ~2.5k cams/node) meshed via `src/cluster` `lease` - camera bound to nearest PoP by `cluster\_key`; `POST /cluster/failover` re-leases on node loss.

3. Storage Architecture - Tiered Hot / Warm / Cold (S3/Ceph)

| Tier | Store | Retention | Content | Retrieval |

Retention policies department-configurable via `fed\_retention` (per `source\_db`/`category`): e.g., `TRAFFIC 90d`, `CRITICAL\_INFRA 365d`; `RegistryEventType.BULK\_IMPORT` drives policy refresh.

8+4).

4. High Availability & Disaster Recovery - Zero Single Point of Failure

Redundant Cluster (Central Gateways, Streaming Nodes, AI Pipelines)

- **API:** Stateless FastAPI (HPA 10-20 replicas) behind Kong/AWS API GW -> `GET /cluster/health` (db/redis/MediaMTX latency); `credential.helper=manager` JWT rotation.

Automated Failover

Sentinel AI - Gujarat CCTV Intelligence Platform

- Heartbeat miss 90s -> `failover` re-assigns `camera\_leases` to next healthy node; browser HLS auto-reconnects (15-segment window masks gap).

Multi-Site DR

| Site | Role | RPO | RTO |

`alembic` versioned restores + `pg\_basebackup` daily; `FED\_DATABASE\_URL` persisted (same PG) survives API restart (verified `fed\_departments 14` -> `320 perms` after restart).

5. Security & Governance - Encryption, RBAC, MFA, Audit for Inter-Department Sharing

| Control | Implementation | Evidence |

****Scale implication:**** RBAC sharded by `cluster\_key`; audit partitioned by `district\_code` (warm) + `fed\_audit` cold tier; 80k cameras add ~800 perms (10 per district) without schema change.

Verification (Baseline Already Proves Scale Contracts)

- `287 pytest pass` + `demo\_scenario 23 pass/0 fail` + `tsc --noEmit` clean